

Exp No: 12

EMBEDDED C PROGRAM TO MULTIPLY TWO 16-BIT BINARY NUMBERS

AIM:

To write an Embedded C program to multiply two 16-bit numbers using Keil software

SOFTWARE REQUIRED:

- Keil

Program:

```
#include <reg51.h>

void main()
{
    unsigned int num1, num2;
    unsigned long int product;
    while (1)
    {
        num1 = 0x2222;
        num2 = 0BBBB;
        product = (unsigned long int)num1 * (unsigned long int)num2;
        P0 = (unsigned char)(product & 0xFF);    // Least significant byte
        P1 = (unsigned char)((product >> 8) & 0xFF); // Second byte
        P2 = (unsigned char)((product >> 16) & 0xFF); // Third byte
        P3 = (unsigned char)((product >> 24) & 0xFF); // Most significant byte
    }
}
```

OUTPUT

